University of Glasgow

ENGLISH LANGUAGE & LINGUISTICS 2A

The Story of Linguistics

The Machine Turn

distributional meaning, word vectors and language models, 1950s to now

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Lecture 3 of 3

A short story

“The dog eyed the birthday cake on the counter. Temptation was calling to him. ‘Stay down,’ he had been told, but the frosting beckoned irresistibly. When nobody was looking, opportunity struck. By the time we returned, guilt was written all over his face. The evidence had been eaten.”



A route through twentieth-century linguistics

I The Scientific Turn

- Saussure (1900s): a system of differences
- Chomsky (1950s–60s): mental computation
- Halliday (1960s): a functional resource
- Hymes (1960s–): communicative competence

II The Cognitive Turn

- Lakoff (1980s): conceptual metaphor
- Langacker (1980s): construal
- Kuno (1970s): empathy
- Goldberg (1990s): constructions

III The Machine Turn

- Shannon (1951): prediction
- Firth and Harris (1950s): distribution
- Word vectors (2013–)
- Transformers (2017–)

What is the next word?

“She takes her coffee with milk and ____”

“The ceremony was held at the local parish ____”

“The evidence had been ____”

Claude Shannon

‘Prediction and entropy of printed English’ (1951)

The experiment

- A human subject guesses a text letter by letter
- After each wrong guess, the subject guesses again
- The number of guesses needed measures how predictable English is

Key concepts

- Entropy: the average information per symbol, in bits
- Redundancy: the part of a text predictable from what came before
- Result: about 1 bit per letter with long contexts, against 4.7 bits for 27 equally likely symbols

What the next word depends on

“The evidence had been ____”

Knowledge	Constraint on the gap
Syntax	“had been” requires a past participle or an adjective
Semantics	evidence can be gathered, destroyed or eaten, but not frightened
The story	the cake has gone, and the dog was alone with it
Register	“evidence” sets up the mock-forensic frame of a crime report

Distributional meaning

from Lecture I

> “…difference of meaning correlates with difference of distribution.”
>
> Harris, ‘Distributional structure’ (1954: 156)

> “You shall know a word by the company it keeps!”
>
> Firth, ‘A synopsis of linguistic theory, 1930–1955’ (1957: 11)

- Distributional hypothesis: words that occur in similar contexts tend to have similar meanings

Collocation

counting the company

Key concepts

- Collocation: words that co-occur more often than chance predicts
- Pointwise mutual information (PMI): how much more often two words occur together than if they were independent
- Church & Hanks (1990): mutual information as a lexicographer's tool

Halliday's example (1966)

strong tea · ?powerful tea

powerful car · ?strong car

Near-synonyms with different company

$$\mathrm{PMI}(x, y) = \log_2 [\, P(x, y) \div (P(x) \times P(y)) \,]$$

From concordance to vector



Data

- Corpus: English Wikipedia, about 900 million words
- Window: five words either side
- Occurrences: frosting 132; icing 842; guilt 5,918; remorse 1,311
- Cosine similarity: frosting–icing 0.70; guilt–remorse 0.79; frosting–guilt 0.02

Latent Semantic Analysis and the TOEFL

Landauer & Dumais 1997

64.4%

LSA: a word-by-document co-occurrence matrix, compressed to about 300 dimensions

64.5%

average score of applicants to US colleges from non-English-speaking countries

Test: 80 synonym items from the Test of English as a Foreign Language

Word vectors

word2vec (Mikolov et al. 2013)

Key concepts

- Word vector (embedding): a list of a few hundred numbers representing a word
- Training: predict the words around each word, over billions of words of text
- Words used in similar contexts end up with similar vectors

The vectors in today's demonstrations

- GloVe (Pennington et al. 2014): vectors from global co-occurrence counts
- 50 numbers per word
- Trained on Wikipedia and the Gigaword news corpus, 6 billion words

The semantic differential

Osgood et al. 1957

The method

- Ratings: a word is rated on seven-point scales between opposite adjectives
- Factors: evaluation (good–bad), potency (strong–weak) and activity (active–passive)
- Result: each word is a point in a three-dimensional space

Rate these words, 1 to 7

Scale	dog	cake
good – bad		
strong – weak		
active – passive		

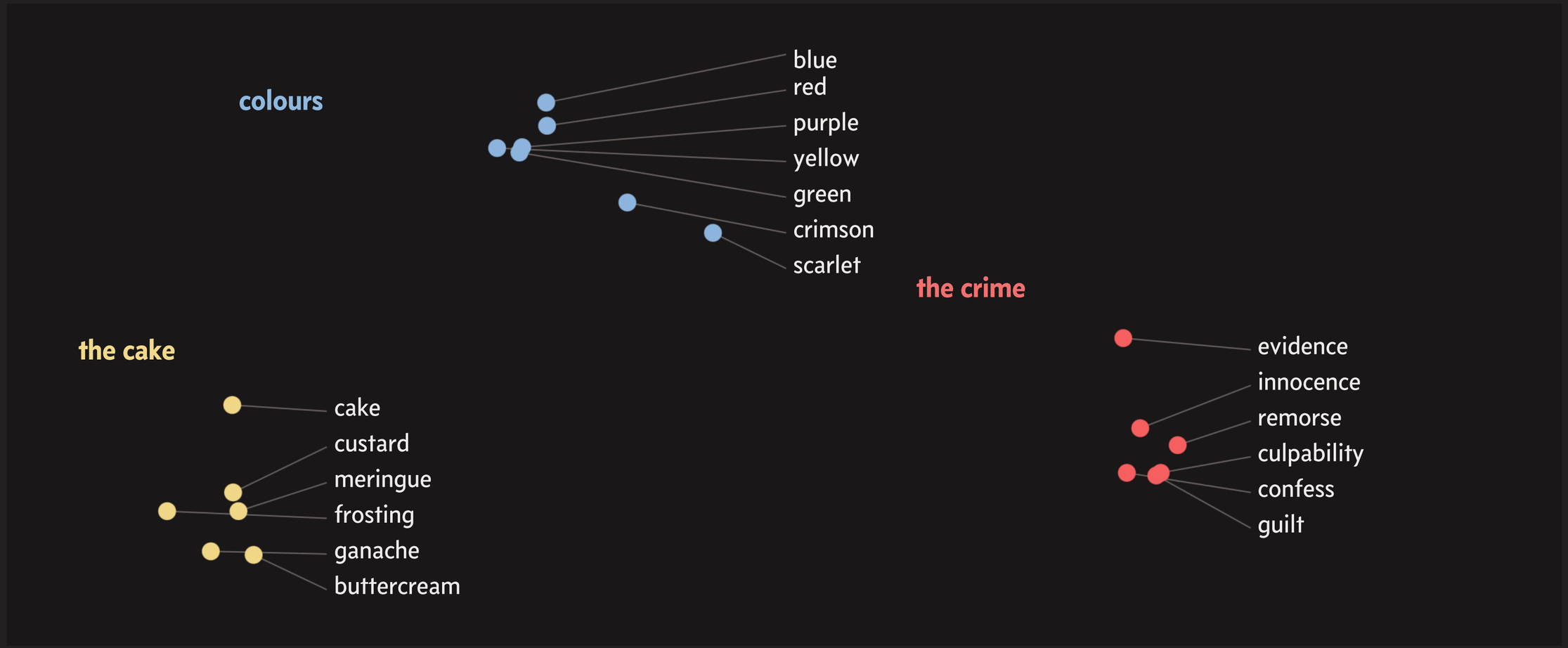
From features to word vectors



Two kinds of feature

- Componential analysis: features chosen by a linguist, such as HUMAN, MALE, ADULT (Katz & Fodor 1963)
- GloVe: 50 unnamed features per word, learned from co-occurrence counts
- king – man + woman: nearest word queen, cosine 0.85

GloVe word vectors in two dimensions



Directions in the vector space



Calculations

- king – man + woman: queen 0.85
- paris – france + scotland: glasgow 0.850, dublin 0.849, edinburgh 0.827
- The maps: flat views through the 50-dimensional space; nearest words are computed in all 50

Nearest neighbours of the passage's words

Word	Five nearest words in GloVe, closest first
frosting	ganache, meringue, custard, caramel, buttercream
guilt	remorse, innocence, betrayal, truth, culpability
temptation	resist, inevitably, vengeance, provoke, tempted
evidence	findings, reveal, case, testimony, revealed

Word vectors and collocation

Levy & Goldberg 2014

1 · counts	2 · association	3 · compression
how often each word occurs near each other word	a word-by-context table of PMI scores	a few hundred numbers per word

- Result: word2vec's skip-gram training implicitly factorises a word–context matrix of PMI values, shifted by a constant
- The analogies appear at step 3: compression keeps the structure the words share

Saussure's *value* and the vector space

Saussure (1916)

- Value: a sign's meaning comes from its differences from other signs in the system
- The Russian blues: синий and голубой divide the space that English "blue" covers

Word vectors (2013–)

- A vector's individual numbers mean nothing on their own
- Rotate the whole space and every distance and direction is unchanged
- Only the relations between vectors carry information

> "…in language there are only differences without positive terms."

Saussure, *Course in general linguistics* (1959 [1916]: 120)

Semantic change in vector space

Hamilton et al. 2016

Word	Neighbours, earlier decades	Neighbours, 1990s
gay	cheerful, frolicsome, witty	homosexual, lesbian, bisexual
broadcast	sow, seed, circulated	television, radio, bbc
awful	solemn, majestic, dread	weird, horrible, terrible

- Method: separate vectors for each decade, 1800–1990, aligned across decades
- Law of conformity: frequent words change meaning more slowly
- Law of innovation: more polysemous words change meaning faster

Bias in word vectors

Bolukbasi et al. (2016)

man : computer programmer :: woman : homemaker

- An analogy produced by word2vec vectors trained on Google News text

Caliskan et al. (2017)

- Implicit Association Test results replicated in GloVe vectors
- Flowers with pleasant words, insects with unpleasant
- Female names with family, male names with career
- European-American names with pleasant words, African-American names with unpleasant

One vector per word

Word	Five nearest words in GloVe
counter	anti, crack, tactics, aimed, action
bank	banks, securities, banking, investment, exchange

- Similarity of counter to attack: 0.65; to kitchen: 0.31
- Type: the word in the dictionary, one vector for all its uses
- Token: the word in a particular sentence, where context selects the sense

From tokens to vectors

the embedding table



GPT-2 small

- Vocabulary: 50,257 tokens, words and word-pieces
- Token id: the token's row in the table; dog is 3290, counter 3753
- Embedding: 768 numbers per token, the same row in every sentence

What does “it” refer to?

“The dog wanted the cake because it was hungry.”

“The dog wanted the cake because it was delicious.”

The transformer

‘Attention is all you need’ (Vaswani et al. 2017)

Key concepts

- Attention: each token’s vector draws information from other tokens’ vectors and is updated
- Attention weight: how much one token draws on another; each token’s weights sum to 1
- Head: one attention pattern, learned in training

GPT-2 small (Radford et al. 2019)

- 12 layers × 12 heads = 144 attention heads
- 124 million parameters
- Each token attends only to itself and earlier tokens
- (This is the little local model I used for this lecture’s animations and figures, and is incredibly small compared to modern LLMs; I would estimate 80+ layers × 128 heads for 10k+ heads across 1 trillion parameters)

Attention on the passage

layer 2, head 7

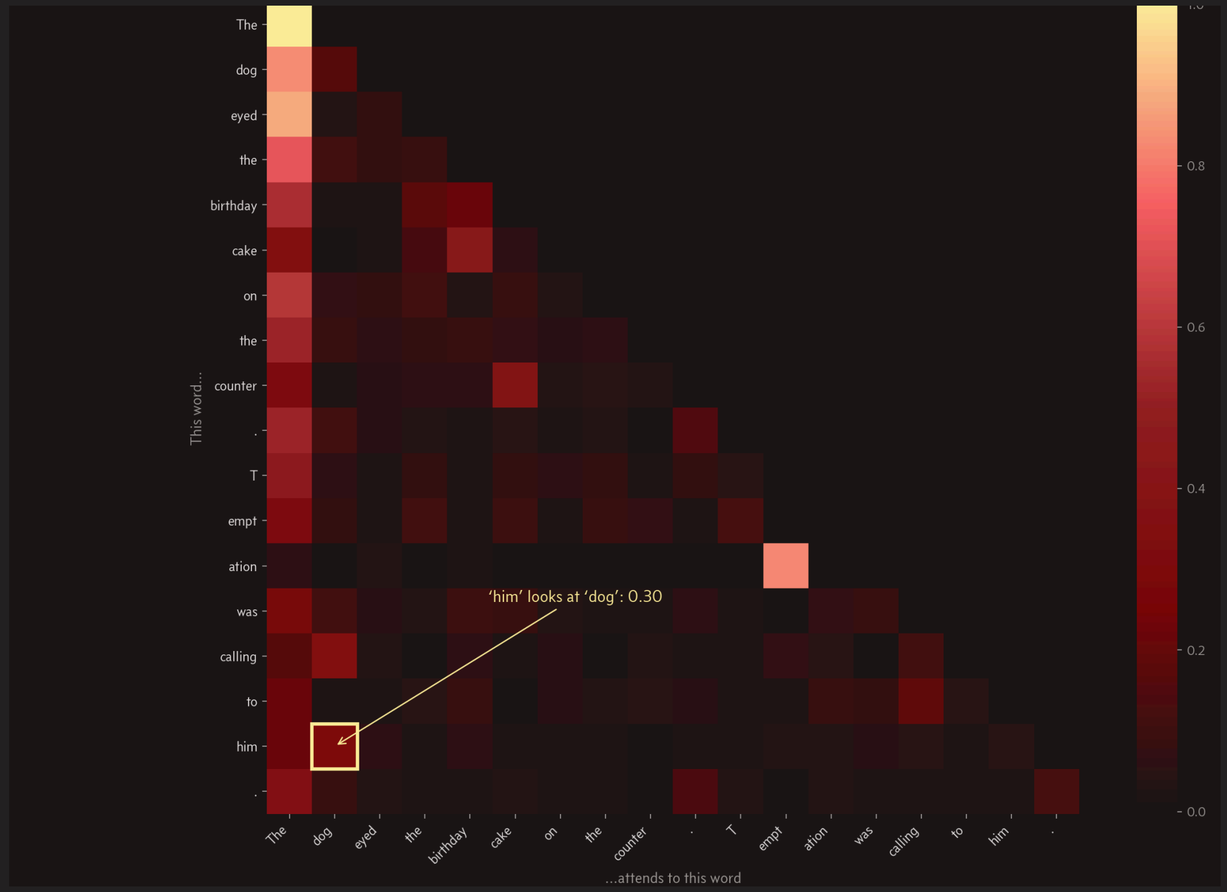


Attention weights

- him → dog: 0.30
- counter → cake: 0.39
- ation → empt: 0.83, within “Temptation”
- Grey arcs: weight on the first token
- Arc width: the weight; each token’s weights sum to 1

One attention head on the passage

- GPT-2 small, layer 2, head 7
- Rows: the attending token
- Columns: the token attended to
- him → dog: 0.30
- Tokens: “Temptation” is split into T + empt + ation



One word, two referents

GPT-2 layer 6, head 11

Sentence	to “dog”	to “cake”
“…because it was hungry.”	0.34	0.10
“…because it was delicious.”	0.06	0.41

- Attention measured from the adjective, since everything before it is identical in the two sentences
- The clearest of GPT-2’s 144 heads, found by inspecting them all

One word, many tokens

counter in GPT-2

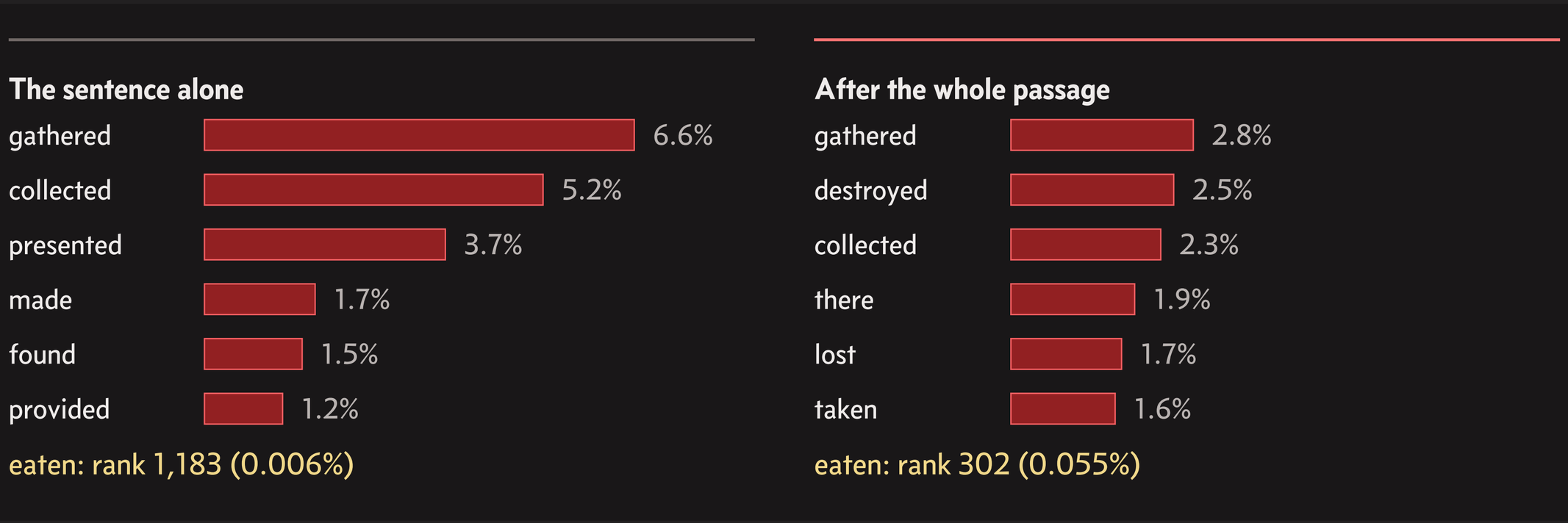


Fifteen sentences

- Senses: a surface, a device and the verb, five sentences each
- Lookup: one identical vector for every token
- Layers 4–9: three separate groups
- Layer 12: the two noun senses move closer
- Map: two of 768 dimensions, about 60% of the variation

“The evidence had been ___”

GPT-2’s next-word probabilities



Generating text

1 · PREDICT

a probability for every possible next token

2 · SAMPLE

choose one token, weighted by probability

3 · APPEND AND REPEAT

the text so far, plus the new token

- Temperature: a setting that sharpens or flattens the probabilities before sampling
- Temperature 0: always the most probable token, so GPT-2 continues “The evidence had been gathered”
- Higher temperatures: less probable tokens are chosen more often
- Chat models: the same loop, after further training on instructions and human feedback

What would each of them say about a language model?

Saussure

a system of differences

Chomsky

an innate grammar

Halliday

language in social context

Hymes

communicative competence

Lakoff

metaphor grounded in the body

Langacker

construal and vantage point

Kuno

empathy and the camera

Goldberg

constructions learned from use

Chomsky and the machine

grammar and probability

> “…in any statistical model for grammaticalness, these sentences will be ruled out on identical grounds as equally ‘remote’ from English.”

Chomsky, *Syntactic structures* (1957: 16), on “Colorless green ideas sleep furiously” and “Furiously sleep ideas green colorless”

≈ 200,000×

Pereira (2000): a simple statistical model trained on newspaper text rates the grammatical order about 200,000 times more probable than the reverse

Chomsky and the machine

the poverty of the stimulus

> “The rise and success of large language models undermines virtually every strong claim for the innateness of language that has been proposed by generative linguistics.”

Piantadosi, ‘Modern language models refute Chomsky’s approach to language’ (2024)

> “…these programs learn humanly possible and humanly impossible languages with equal facility.”

Chomsky, Roberts & Watumull, *The New York Times* (8 March 2023)

Chomsky and the machine

impossible languages (Kallini et al. 2024)

The experiment

- GPT-2 models trained on English and on altered versions of English
- Word order shuffled at random
- Sentences reversed
- Grammatical markers placed by counting words, a rule no human language uses

The result

- The models learned the impossible languages less well than English
- The more unnatural the language, the worse the learning
- Best paper award, ACL 2024

Halliday and the machine

register

Variable	Definition	Supplied in the prompt
Field	what is going on	“the dog and the birthday cake”
Tenor	who is taking part, and their relationship	“from the vet to the owner”
Mode	the part language plays	“a written incident report”

Ask an LLM for the passage as a vet’s incident report, then as a WhatsApp message, then as a Shakespearean sonnet, etc. This is the 2023 party trick.

Hymes and the machine

speech acts

A model writes: “I promise to remind you tomorrow.”

Condition	For a promise (Searle 1969)
Preparatory	the hearer wants it done; it would not happen anyway
Sincerity	the speaker intends to do it
Essential	saying it places the speaker under an obligation

Communicative competence: knowing what is appropriate, to whom, when, and with what standing (Hymes 1972)

Lakoff and the machine

embodiment

The claim (Lakoff & Johnson 1980)

- Basic metaphors come from correlations in bodily experience

MORE IS UP

piles grow taller as quantity increases

HAPPY IS UP

posture: drooping with sadness, upright with joy

The training data

- GPT-2: text only, about 8 million web pages (40 GB)
- Current frontier models: text, images and audio
- No model has a body or a posture
- All of them use and explain these metaphors

Langacker, Kuno and the machine

viewpoint on request

Request to a model	The theory's terms
"Tell it from the humans' point of view"	Kuno: camera with the humans
"Tell it from the cake's point of view"	Kuno: camera with an inanimate object
"Take the narrator out of the story"	Langacker: objective construal, no "we" on stage

Goldberg and the machine

learning a rare construction

“a beautiful five days” “*a five beautiful days”

- The AANN construction: article + adjective + numeral + plural noun
- Training: 100 million words, with every AANN example removed
- Result: models still preferred the AANN to corrupted versions
- Learned from: “a few days”, “five miles is a long way”

Misra & Mahowald (2024); Mahowald (2023)

Eight approaches and the language model

Approach	Core claim	Evidence from language models
Saussure	value from differences	meaning as relative position in a vector space
Chomsky	grammar is not learnable from statistics	Pereira's 200,000×; impossible languages learned less well
Halliday	register: field, tenor, mode	any register, with the situation supplied by the prompt
Hymes	appropriateness and standing	appropriate-sounding text; felicity conditions in question
Lakoff	metaphor from bodily experience	embodied metaphors used fluently, without a body
Langacker	construal and vantage point	any vantage point on request
Kuno	empathy and the camera	any camera angle on request
Goldberg	constructions learned from input	a rare construction learned from related ones

Work for linguists

Most of language

- About 7,000 living languages
- Most have little or no text online
- Documentation, description and revitalisation

Whose English?

- Training text over-represents some varieties of English
- Auditing models across varieties, registers and dialects

New fieldwork

- Interpretability: finding out what models have learned about language
- Models as objects of linguistic study

From GPT-2 to the frontier, 2019–2026

GPT-2 (OpenAI, 2019)

- Every demonstration in today's lecture
- 124 million parameters (small) to 1.5 billion (largest)
- Open: anyone can inspect its weights and attention
- Ranks "eaten" 302nd after the whole passage

Frontier models (2026)

- The models you may be most likely to use, if you choose to use AI models
- Sizes mostly undisclosed, parameters likely in the 1 trillion range
- Pass professional examinations; converse across many languages
- Explain who ate the cake, and why the passage is funny

Emotion concepts in a language model

2026

The finding

- Model studied: Claude Sonnet 4.5
- Patterns for 171 emotion concepts; similar emotions, similar patterns
- “Desperate” rises as the model fails a task
- Boosting it raises cheating and blackmail

The term

- Functional emotion: shapes behaviour as an emotion would
- Steering: artificially boosting a pattern to test it
- No trace: boosted desperation can drive cheating in calm-sounding text

> “Note that none of this tells us whether language models actually feel anything or have subjective experiences.”

Anthropic, ‘Emotion concepts and their function in a large language model’ (2 April 2026)

The sceptics

form without meaning

> "…an LM is a system for haphazardly stitching together sequences of linguistic forms it has observed in its vast training data, according to probabilistic information about how they combine, but without any reference to meaning: a stochastic parrot."
>
> Bender, Gebru, McMillan-Major & Shmitchell (2021: 616–617)

> "…a system trained only on form has a priori no way to learn meaning."
>
> Bender & Koller (2020: 5185)

Meaning: for Bender & Koller, the relation between a form and something outside language, such as the world or a speaker's intent

The passage, one last time

“The dog eyed the birthday cake on the counter. Temptation was calling to him. ‘Stay down,’ he had been told, but the frosting beckoned irresistibly. When nobody was looking, opportunity struck. By the time we returned, guilt was written all over his face. The evidence had been eaten.”



A century of linguistics

objects of study

Approach	Object of study
Saussure (1916)	the language system
Chomsky (1957–)	the speaker’s mental grammar
Halliday, Hymes (1960s–70s)	language in social context
Lakoff, Langacker, Kuno, Goldberg (1970s–90s)	thought, perspective and learning
Language models (2013–)	distribution at scale

Credits

Animations

- Made for this lecture with `manimgl`, Grant Sanderson's open-source animation library (see 3Blue1Brown)
- Vectors drawn as colour strips, after Alammar (2019)
- Values computed from GloVe vectors, Wikipedia counts, GPT-2 weights and states

Data

- Word vectors: GloVe, glove-wiki-gigaword-50 (Pennington et al. 2014)
- Attention and probabilities: GPT-2 small (Radford et al. 2019)
- Concordances and counts: English Wikipedia, November 2023, about 900 million words

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